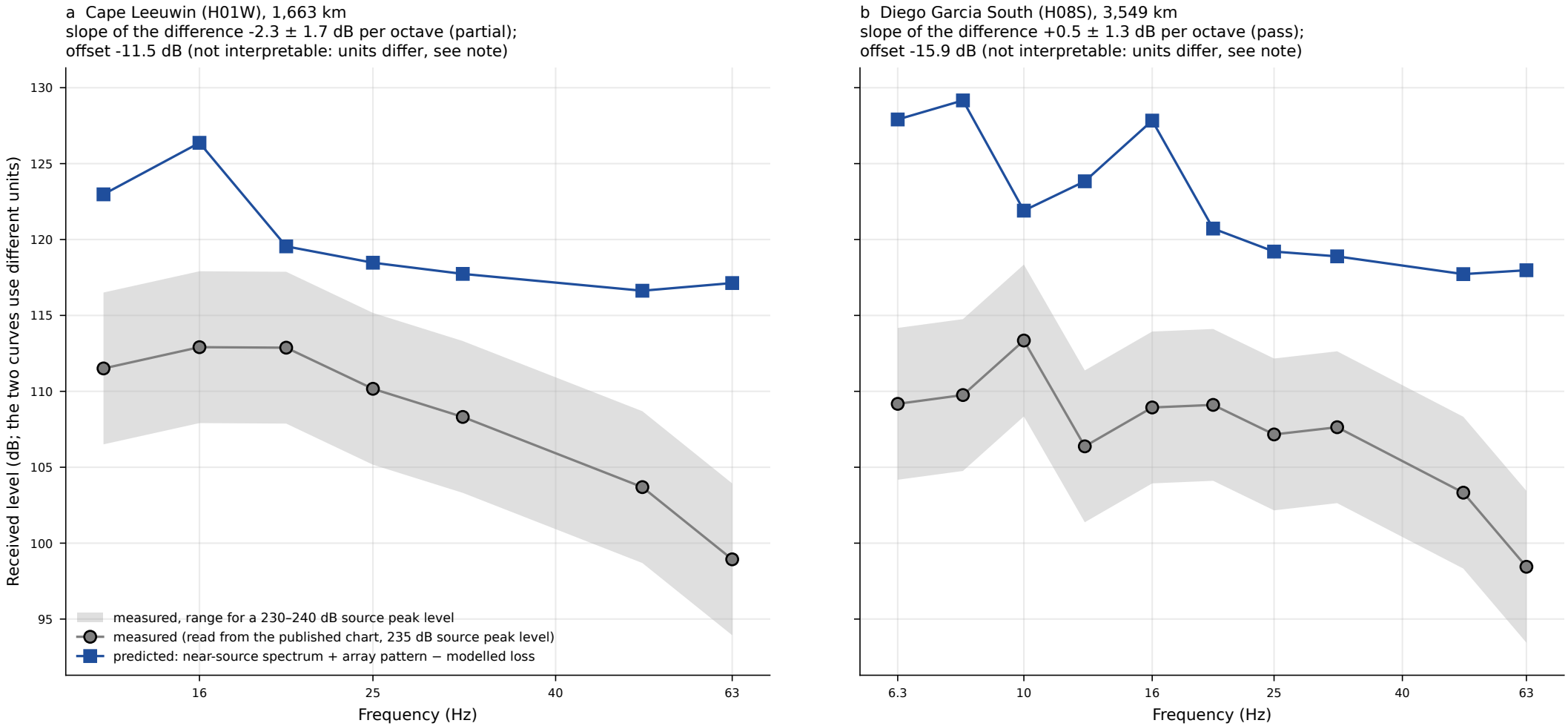
Airgun shots of October 2001 (site air9): predicted against measured received level at two listening stations,
with the source spectrum from the ship's near-source recording — verdict NOT EXPLAINED (units of the level test, gap G23)



Plain note: For one set of airgun shots fired in 2001, the chart compares the sound level that our model predicts at two listening stations (blue) with the level that was measured (grey). The model takes the source sound from a recording made near the airguns. The change of level with frequency now agrees at both stations. The two curves are not at the same height, but the measured value is a peak level and the predicted value is a spectrum level, so the difference cannot be read as a model error. The rule fixed before the test gives the verdict NOT EXPLAINED. The frequency caveat C1 stays in force.

Technical note: RL_pred = S0 + AF − TL; S0 = EW0112 Fig. 2 Ball spectrum (DIGITISED, two shots, dB mean) + 20 log10(25.75 m) − near-field reference factor (OFR 95-228 20-gun layout, V^(1/3) weights, Ball 30 m ahead, 15 m deep, source 10 m); AF = horizontal array factor toward the station; TL = KRAKEN adiabatic incoherent modes, hard bottom, Francois-Garrison, GEBCO_2026/WOA23 (Oct) sections, receiver at triad depth. RL_obs = SL_B − TL_obs (Blackman et al. 2004 Fig. 23, DIGITISED +/- 2 dB), SL_B = 235 dB (band 230–240). Bands: H01W 12.5–63 Hz (7), H08S 6.3–63 Hz (10); 40 Hz excluded. Grades (audit criteria 940f1a3), nominal geometry: H01W slope of the residual -2.3 +/- 1.7 dB/oct (PARTIAL), level -11.5 dB (FAIL), shape 3.1 dB (PARTIAL); H08S slope +0.5 +/- 1.3 dB/oct (PASS), level -15.9 dB (FAIL), shape 3.9 dB (PARTIAL); common mode after the EW0112 source -1.2 +/- 1.8 dB/oct (about −9 with a flat source); slope PARTIAL-or-better at both stations on 0.78 of 45 geometries. Verdict under pre-registration e949144: NOT EXPLAINED - the level test fails for every SL_B in 230–240 dB, and that test is not interpretable because the measured side is a peak level and the predicted side a spectrum level (units mismatch of unknown size: gap G23, declared; no re-grade). Training case, not a blind test; the independent-site test was NOT TESTABLE. C1 unchanged: propagation slope misfit +10.4 / +7.4 dB per octave (uncorrected, H01W / H08S); air9 absolute loss NOT CALIBRATED in slope for any likelihood. No MH370 impact posterior enters this chart.